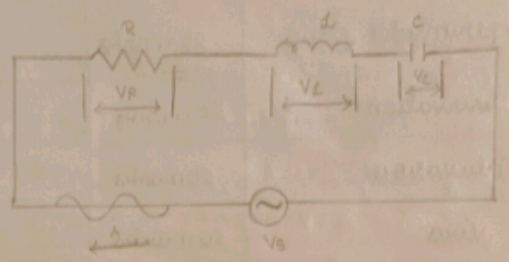
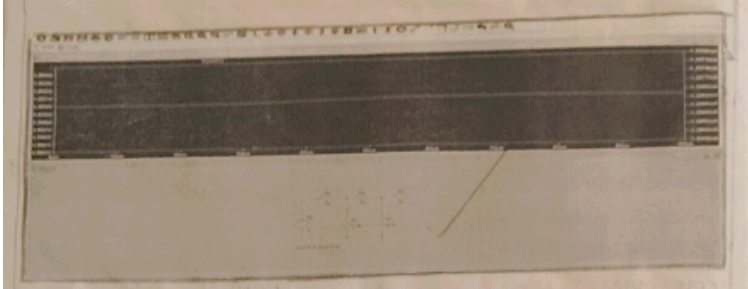
CIRCUIT DIAGRAM:

where R = 10kΩ, L = 10mH and C = 0.1μF (Values of R, L and C can be changed)





Ex. NO: 08	TRANSIENT ANAYSIS OF SERIES RLC CIRCUIT
Date: 06/03/2025	

AIM:

To determine transient response of a series RLC circuit using LTspice simulator.

APPARATUS REQUIRED:

SNO	COMPONENTS	SPECIFICATIONS	QUANTITY
1.	CRO	10MHz	1
2.	DCB, DIB	---	1
3.	Resistor	10kΩ	1
4.	Bread board	---	1
5.	Connecting wires	---	---
6.	PC with LTspice simulator	---	---

PROCEDURE:

i) Make connections as per the circuit diagram.

ii) Calculate the values of $V_C(t)$, $V_R(t)$ and write in table. Compare the calculated values and measured values of voltage in both the cases.

TABULAR COLUMN:

Voltage across resistor:

TIME (t)	$V_R(t)$ - VOLTAGE ACROSS RESISTOR OBSERVED VALUE
$t=0$	1.46V
$t=\tau=1ms$	6.34V
$t=2\tau=2ms$	8.66V
$t=3\tau=3ms$	9.44V
$t=4\tau=4ms$	9.81V
$t=5\tau=5ms$	9.855V

MARKS SPLIT-UP

S.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	Experimental setup	20 marks	20
2.	Observation	20 marks	20
3.	Calculation	20 marks	20
4.	Viva	20 marks	18
5.	Record	20 marks	20
			98

RESULT:

Thus the transient analysis of series RLC circuit is done practically using LTspice simulator.